Statistical Data Analysis

Lecture 11: Linear regression analysis, part I

Marc Corstanje

Periods 4 & 5

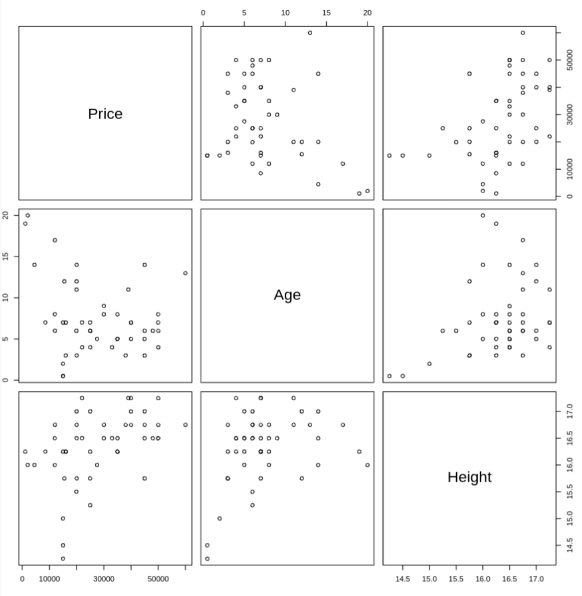
Where Are We?

- Chapter 7: Analysis of categorical data
  - Still on May 13th: discussion of Assignment 6 & Quiz 6
  - Deadline is on May 6th
- Chapter 8: Linear regression analysis
  - Section 8.1 – The multiple linear regression model
  - Section 8.2 – Diagnostics
  - Section 8.3 – Collinearity

No class on April 29th and no practicals on May 1st

Multiple Linear Regression

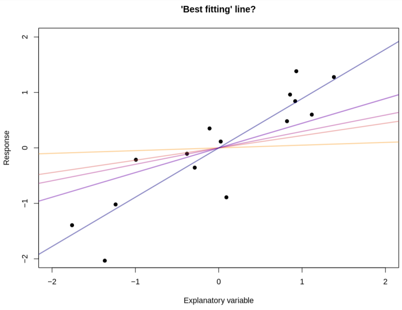
What Drives the Price of Horses? 🐴



What is Linear Regression?

- Goal of regression: model response (dependent) variable given explanatory (independent) variables

$$\text{response variable} = f(\text{explanatory variables}) + \text{measurement error}$$



- $f \rightsquigarrow$ systematic info that explanatory variables provide about response
- f assumed linear in linear regression
- f unknown $\rightsquigarrow$ needs to be estimated for prediction and inference

- We consider multiple linear regression
  - One response variable and more than one explanatory variable

Linear regression model

- Input
  - Response variable Y
  - Explanatory variable x
  - Error e
- Linear regression equation

$$Y = f_\beta(\boldsymbol{x}) + e,$$

where $e \sim N(0, \sigma^2)$ and

$$f_\beta(\boldsymbol{x}) = \beta_0 + x_1\beta_1 + \cdots + x_p\beta_p$$

- Observed data
  - inputs $\boldsymbol{x}_1, \ldots, \boldsymbol{x}_n$, where $\boldsymbol{x}_i = (x_{i1}, \ldots, x_{ip})$
  - Realizations $y_1, \ldots, y_n$ of $f_\beta(\boldsymbol{x}_i) + e$, from which we derive realizations $e_1, \ldots, e_n$ of e.
- Goal: Estimate β based on the observation $(y_1, \boldsymbol{x}_1), \ldots, (y_n, \boldsymbol{x}_n)$.

Notation

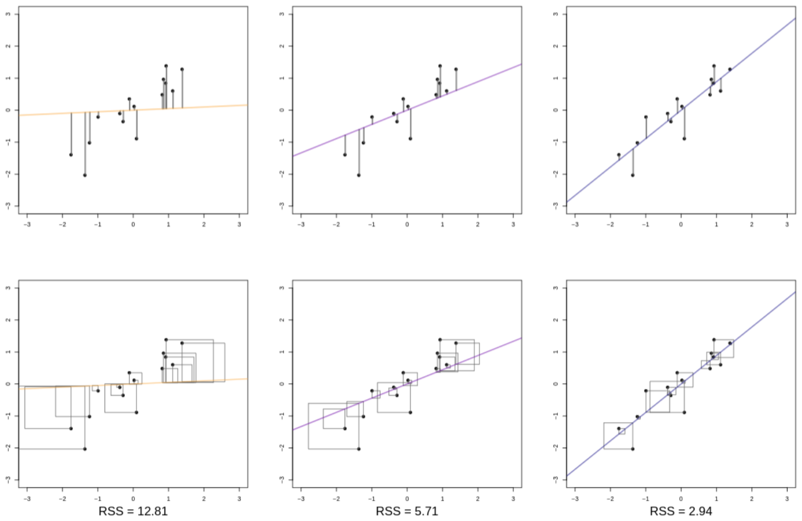
- n observations and $p < n$ explanatory variables
  - Indices for observational units: $i = 1, \dots, n$
  - Indices for explanatory variables: $j = 1, \dots, p$

	'Scalar' notation	Matrix notation
Model	$Y_i = \beta_0 + x_{i1}\beta_1 + \cdots + x_{ip}\beta_p + e_i$	$\boldsymbol{Y} = \boldsymbol{X}\boldsymbol{\beta} + \boldsymbol{e}$
Response	Y_i (realization: y_i)	$\boldsymbol{Y} = (Y_1, \dots, Y_n)^T$
Explanatory Variables	x_{ij}	$\boldsymbol{X} = \begin{pmatrix} 1 & x_{11} & \cdots & x_{1p} \\ \vdots & \vdots & & \vdots \\ 1 & x_{n1} & \cdots & x_{np} \end{pmatrix}$
Coefficients	$\beta_0, \beta_1, \dots, \beta_p$	$\boldsymbol{\beta} = (\beta_0, \beta_1, \dots, \beta_p)^T$
Assumptions on Errors	$\mathbb{E}(e_i) = 0, \; \mathbb{E}(e_i e_j) = \begin{cases} \sigma^2, & i = j, \\ 0, & i \neq j \end{cases}$	$\mathbb{E}(\boldsymbol{e}) = \boldsymbol{0}, \; \mathrm{Cov}(\boldsymbol{e}) = \sigma^2 \mathbb{I}_{n \times n}$

- Further assumptions
  - Full-rank design matrix, i.e. $\mathrm{rank}(\boldsymbol{X}) = p + 1$
  - $e_i \sim N(0, \sigma^2)$, so that $Y \mid \boldsymbol{X} = \boldsymbol{x} \sim N\left(\beta_0 + x_1\beta_1 + \cdots + x_p\beta_p, \sigma^2\right)$

Model Fitting

Parameter Estimation via OLS – Illustration



? What are the least squares parameter estimates $\hat{\beta}_0$ and $\hat{\beta}_1$ in simple linear regression?

? And in multiple linear regression?

Parameter Estimation via OLS – Estimators & Residuals

- Note, $y_i - f_{\beta}(\boldsymbol{x}_i)$ are realizations of $e \sim N(0, \sigma)^2$. The likelihood

$$L(\beta) = \prod_{i=1}^{n} \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(y_i - f_{\beta}(\boldsymbol{x}_i))^2}{2\sigma^2}\right)$$

  is maximized when $\sum_{i=1}^{n} (y_i - f_{\beta}(\boldsymbol{x}_i))^2$ is minimized.

- Least squares: minimize $S(\beta) = (\boldsymbol{Y} - \boldsymbol{X}\beta)^T(\boldsymbol{Y} - \boldsymbol{X}\beta) = \|\boldsymbol{Y} - \boldsymbol{X}\beta\|^2$
  - Coefficient vector estimated by $\hat{\beta} = (\boldsymbol{X}^T\boldsymbol{X})^{-1}\boldsymbol{X}^T\boldsymbol{Y}$
  - Unbiased, moreover $\mathrm{Cov}(\hat{\beta}) = \sigma^2(\boldsymbol{X}^T\boldsymbol{X})^{-1}$
  - Predicted response vector: $\hat{\boldsymbol{Y}} = \boldsymbol{X}\hat{\beta}$, where $\hat{Y}_i = \hat{\beta}_0 + x_{i1}\hat{\beta}_1 + \cdots + x_{ip}\hat{\beta}_p$
  - Residuals: $R_i = Y_i - \hat{Y}_i$, $i = 1, \ldots, n$

- Residual sum of squares: $\mathrm{RSS} = S(\hat{\beta}) = \sum_{i=1}^{n}(Y_i - \hat{Y}_i)^2 = \|\boldsymbol{Y} - \boldsymbol{X}\hat{\beta}\|^2$
  - Error variance estimated by $\hat{\sigma}^2 = \frac{RSS}{n-p-1}$
  - Then, $\widehat{\mathrm{Cov}}(\hat{\beta}) = \hat{\sigma}^2(\boldsymbol{X}^T\boldsymbol{X})^{-1}$

Parameter Estimation via OLS – R Output

```
library(Stat2Data); data("HorsePrices")
# Remove rows with NAs and exclude variables (columns) ID and Sex
horses <- HorsePrices[complete.cases(HorsePrices), 2:4]
class(horses)
```

```
## [1] "data.frame"
```

```
### Linear model syntax: lm(formula, data, ...)
# First argument is of class 'formula': response ~ first_var + second_var + ...
# The (optional) 'data' should be (coercible to) a data.frame
horses_lm <- lm(Price ~ Age + Height, data = horses)
# The output is an object of class "lm", a summary of which can be printed with:
summary(horses_lm)
```

```
##
## Call:
## lm(formula = Price ~ Age + Height, data = horses)
##
## Residuals:
##    Min     1Q Median     3Q    Max
## -22842  -8606   -180   7360  34511
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) -172210.0    45285.9  -3.803 0.000437 ***
## Age           -1432.8      418.6  -3.423 0.001352 **
## Height        12915.0     2824.9   4.572 3.91e-05 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 12010 on 44 degrees of freedom
## Multiple R-squared:  0.3653, Adjusted R-squared:  0.3365
## F-statistic: 12.66 on 2 and 44 DF,  p-value: 4.529e-05
```

Variable Selection

Variable Selection – Idea

- Problem: Too many explanatory variables
  - More variables will trivially improve the fitting (reduce RSS)
  - But the dimension can become infeasible
  - Risk of overfitting
  - Which variables should we remove?
- Goal: obtain the best model with the smallest number of explanatory variables
  - ? How to choose the explanatory variables to be included in the model?
  - ? 'Best' according to which criterion?
- Stepwise procedures to include or exclude variables
  - Step forward (or up): start with empty model, add one by one
  - Step backward (or down): start with full model, remove one by one
  - Note: most selection criteria depend on variables already/still in the model $\rightsquigarrow$ many combinations to be considered
  - ? Which criterion of inclusion/removal to choose?

Variable Selection – Coefficient of determination

- Given $\boldsymbol{X}$, we estimate Y by $\hat{y}_i = f_{\hat{\beta}}(\boldsymbol{x}_i) = \boldsymbol{X}\hat{\beta}$.
- For the total variance in the data, we have

$$\mathrm{SSY} = \mathrm{RSS} + \mathrm{SS}_{\mathrm{reg}}$$

Variance caused by the lineair model $\sum(\hat{y}_i - \bar{y})^2$

Variance caused by errors in the regression $\sum(y_i - \hat{y}_i)^2$

Total variance $\sum(y_i - \bar{y})^2$

- RSS is the unexplained variance
- $\mathrm{SS}_{\mathrm{reg}}$ is the explained variantie
- **Coefficient of determination:** $\mathcal{R}^2 = \frac{\mathrm{SS}_{\mathrm{reg}}}{\mathrm{SSY}} = 1 - \frac{\mathrm{RSS}}{\mathrm{SSY}}$

Variable Selection – Coefficient of Determination

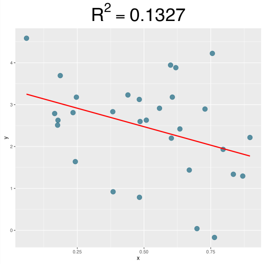
- Global check of model fit: determination coefficient
- Idea: compare given model $\boldsymbol{Y} = \boldsymbol{X}\beta + \boldsymbol{e}$ with empty one $\boldsymbol{Y} = \boldsymbol{1}\beta_0 + \boldsymbol{e}$
  - In the empty model, $\hat{\beta}_0 = \overline{Y}$
  - Thus, the residual sum of squares of the empty model is the total sum of squares $\text{SSY} = \sum_{i=1}^{n}(Y_i - \overline{Y})^2$
- Determination coefficient: $\mathcal{R}^2 = \frac{\text{SSY}-\text{RSS}}{\text{SSY}} = 1 - \frac{\text{RSS}}{\text{SSY}} \rightsquigarrow 0 \leq \mathcal{R}^2 \leq 1$
  - Also called fraction of explained variance
  - High value of $\mathcal{R}^2 \rightsquigarrow$ good fit, but no fixed thresholds

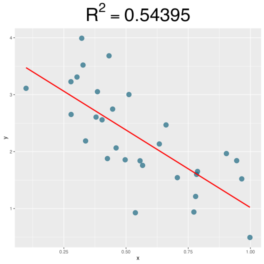
```
## Residual standard error: 12010 on 44 degrees of freedom
## Multiple R-squared:  0.3653, Adjusted R-squared:  0.3365
## F-statistic: 12.66 on 2 and 44 DF,  p-value: 4.529e-05
```

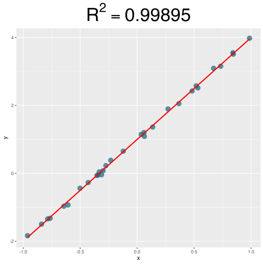
```
# Manually
RSS <- sum(residuals(horses_lm)^2)
SSY <- sum((horses$Price - mean(horses$Price))^2)
(SSY - RSS)/SSY
```

```
## [1] 0.3653362
```

Variable Selection – Coefficient of Determination







- $\mathcal{R}^2$ gives a good indication of the quality of the fit, but
  - No threshold or relative
  - More variables automatically yields a higher $\mathcal{R}^2$, but increase can be minimal

Variable Selection – F-Tests

- *Overall F-test*: test $H_0 : \beta_1 = \beta_2 = \cdots = \beta_p = 0$ vs. $H_1 : \beta_j \neq 0$ for some $j = 1, \ldots, p$
- If errors i.i.d. normal, $F = \frac{(n-p-1)(SSY-RSS)}{p \cdot RSS} \overset{H_0}{\sim} F_{p,n-p-1}$
  - We reject H_0 for large values of F

```
## Residual standard error: 12010 on 44 degrees of freedom
## Multiple R-squared:  0.3653, Adjusted R-squared:  0.3365
## F-statistic: 12.66 on 2 and 44 DF,  p-value: 4.529e-05
```

- *Partial F-test*: with other coefficients arbitrary and for $q > p$, test $H_0 : \beta_{p+1} = \cdots = \beta_q = 0$ vs. $H_1 : \beta_j \neq 0$ for some $j = p+1, \ldots, q$

```
anova(lm(Price ~ Height, data = horses), horses_lm)
```

```
## Analysis of Variance Table
##
## Model 1: Price ~ Height
## Model 2: Price ~ Age + Height
##   Res.Df        RSS Df  Sum of Sq      F   Pr(>F)
## 1     45 8035593114
## 2     44 6346089091  1 1689504023 11.714 0.001352 **
```

Variable Selection – t-Tests

- Test for the inclusion of a single X_k: t-test
  - Note: to address the same problem, considerations about the partial correlation coefficient can also apply
- With β_j arbitrary for $j \neq k$, test $H_0 : \beta_k = 0$ vs. $H_1 : \beta_k \neq 0$
- If errors i.i.d. normal, $T_k = \frac{\hat{\beta}_k}{\sqrt{\widehat{\text{Cov}}(\hat{\beta})_{kk}}} \overset{H_0}{\sim} t_{n-p-1}$
  - Two-sided test
  - Equivalent to partial F-test for the additional inclusion of (only) X_k

```
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) -172210.0    45285.9  -3.803 0.000437 ***
## Age           -1432.8      418.6  -3.423 0.001352 **
## Height        12915.0     2824.9   4.572 3.91e-05 ***
```

Variable Selection – Further Considerations

- Model selection problem: for partial F-tests and t-tests, the inclusion decision may depend on the already present variables
  - E.g., in a step selection procedure, the final model may depend on the order in which variables are progressively tested for inclusion
- Fit improvement caveat: a variable included on the basis of a t- or (partial) F-test may hardly improve the model $\mathcal{R}^2$
  - Note: $\mathcal{R}^2$ never descreases with the inclusion of additional variables
  - We want a model as simple as possible, while as complex as necessary
  - Practical considerations are also relevant for model selection
  - Adjusted $\mathcal{R}^2$ penalizes for the inclusion of unnecessary predictors

Summary and wrap-up

Summary

- In linear regression, we assume a linear dependence between an explanatory variable $\boldsymbol{x}$ and a response variable Y, given by

$$Y = \beta_0 + x_1\beta_1 + \cdots + x_p\beta_p + e,$$

  where
  - The regression coefficients $\beta_0, \ldots, \beta_p$ must be estimated from the pairs $(y_1, \boldsymbol{x}_1), \ldots, (y_n, \boldsymbol{x}_n)$
  - The errors e are independent $N(0, \sigma^2)$-distributed.
- For variable selection, we consider
  - $\mathcal{R}^2 = \frac{\text{explained variance}}{\text{total variance}}$ to assess the quality of the fit
  - The overall F-test to test whether all coefficients are 0
  - Partial F-test to compare models
  - t-tests to test for the inclusion of a single variable

What now?

- Continue assignment 6
  - All material now covered
  - Deadline May 6, 23:59
- Notice: No lecture and tutorial next week